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Machine Learning for Shock-Capturing Numerical Schemes

Learning reconstruction decisions for
1D hyperbolic conservation laws

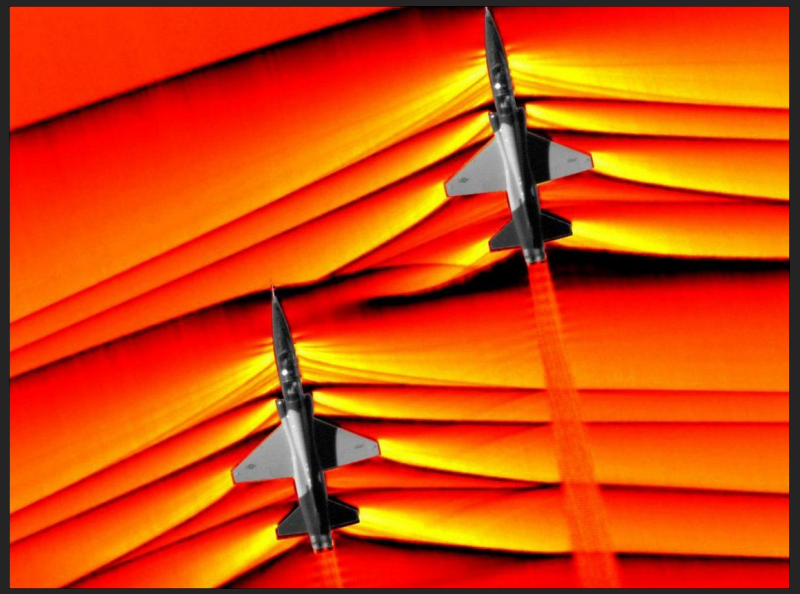
WENO-NN / NN-TENO5 / DIFFERENTIABLE CFD

Yasser Baouzil

Supervisor: Radouan Boukharfane

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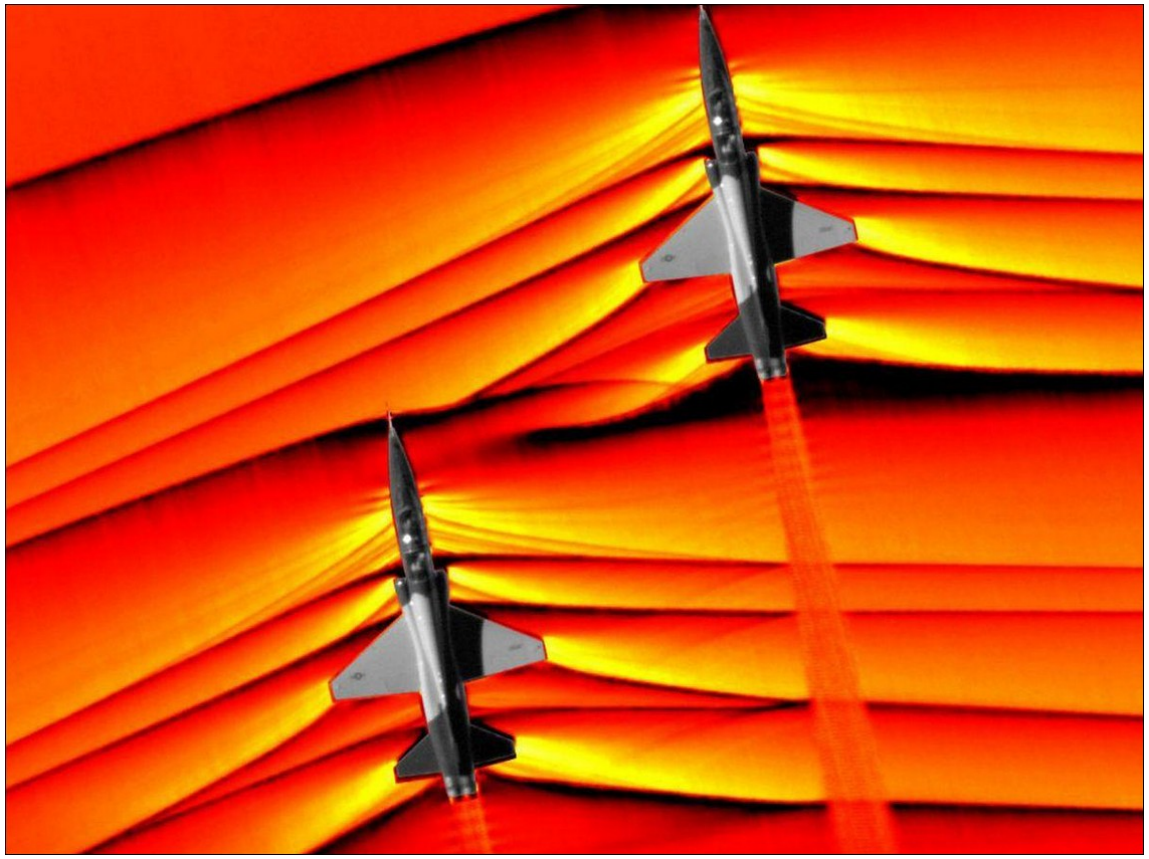
PHYSICS × NUMERICS × LEARNING



Capturing sharp fronts without spurious oscillations

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Can learned reconstruction improve TENO5?



Sharp fronts require both accuracy and control of oscillations.

Research question

Can a small learned correction reduce solution error relative to TENO5, while keeping Euler rollouts stable?

What changes?

Candidate decisions and a learned mixing coefficient.

What stays?

The conservative flux update and SSPRK3 solver.

How is it tested?

Four 1D benchmarks against physical references.

> Aim: improve reconstruction inside the solver—not replace the physical equations.

The solver follows a conservative numerical path

Forward numerical path

State + local stencil → Scalar / Euler setup → **WENO / TENO or NN selector** → Interface flux → SSPRK3 step → K-step rollout

Learned component

Conservation through shared fluxes

$$\frac{d\bar{u}_i}{dt} = -\frac{\hat{f}_{i+1/2} - \hat{f}_{i-1/2}}{\Delta x}$$

Each cell changes by its incoming and outgoing flux.

Follow the path in order

Set up the physical problem.
Reconstruct the interface flux.
Advance with SSPRK3, then repeat K steps.

Detailed slides follow this path: physical setup → reconstruction → fluxes → time stepping.

Euler equations describe compressible flow

The system tracks three conserved quantities throughout the flow.

Density

How mass is distributed in the flow.

Momentum

How the motion of the fluid is transported.

Total energy

Internal and kinetic energy together.

State used by the solver

$$U = (\rho,\ \rho v,\ E)^T$$

ρ: density v: velocity E: total energy

Physical states must remain valid

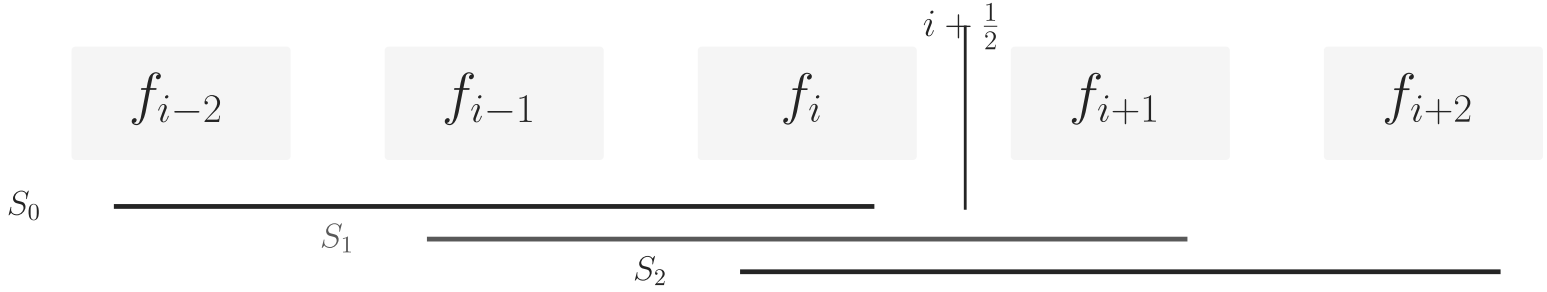
$$\rho > 0, \qquad p > 0$$

Density and pressure must remain positive.

> Sod and Shu–Osher test the learned reconstruction in compressible Euler flows.

WENO and TENO combine three local reconstructions

Five stencil values → three third-order candidates → one interface flux



ENO selects

Choose the smoothest candidate.

Keep one reliable candidate.

WENO weights

Reduce troubled candidates continuously.

Smooth region:
spread trust across the candidates.

TENO discards

Reject troubled candidates, then reweight.

Near a shock:
reject a candidate crossing the jump.

$$\hat{f}_{i+1/2} = \omega_0 q_0 + \omega_1 q_1 + \omega_2 q_2$$

Combine with local weights.

The final model learns decisions and their influence

Two reconstruction paths share the same local data; a small head controls their blend.

Local five-point stencil

Only 4 trainable parameters

↓ ↓

Classical TENO5

Baseline reconstruction

Φ_{TENO5}

Neural reconstruction

Classifier: 5 → 64 → 32 → 16 → 8 → 3

Φ_{NN} Outputs: p_0, p_1, p_2

→

Dedicated mixing head

Three probabilities → sigmoid → λ

$0 \leq \lambda \leq 1$

↓ ↓ ↓

$$\Phi_{\mathrm{final}} = (1-\lambda)\Phi_{\mathrm{TENO5}} + \lambda\Phi_{\mathrm{NN}}$$

↓

Conservative flux update + SSPRK3

Build the interface flux, then advance with RK3

Flux splitting → WENO / TENO or NN reconstruction → interface flux → SSPRK3 → K-step rollout.

01 Flux splitting and interface flux

$$F^{\pm}(U) = \tfrac{1}{2}\left[F(U) \pm aU\right]$$

WENO / TENO or NN reconstructs each split flux.

$$\hat{F}_{i+1/2} = \hat{F}^{+}_{i+1/2} + \hat{F}^{-}_{i+1/2}$$

$$L_i(U) = -\frac{\hat{F}_{i+1/2} - \hat{F}_{i-1/2}}{\Delta x}$$

→

02 SSPRK3 · three stages

$$U^{(1)} = U^n + \Delta t\, L(U^n)$$

Recompute reconstruction and fluxes at each stage.

$$U^{(2)} = \tfrac{3}{4}U^n + \tfrac{1}{4}\left[U^{(1)} + \Delta t L(U^{(1)})\right]$$

$$U^{n+1} = \tfrac{1}{3}U^n + \tfrac{2}{3}\left[U^{(2)} + \Delta t L(U^{(2)})\right]$$

L(U): spatial flux balance. a: wave-speed bound. Each RK stage recomputes reconstruction and fluxes.

03 K-step rollout: repeat the complete SSPRK3 update K times.

Train on the solution after real solver steps

Imitation learns TENO5 labels. Physics training instead optimizes the evolved solution.

Start from a physical state → **Run K complete SSPRK3 steps** → **Compare with the reference**

The training gradient passes through reconstruction, fluxes and every time-integration stage.

$$\mathcal{L}_{\text{total}} = \alpha_1 \mathcal{L}_{\text{MSE}} + \alpha_2 \mathcal{L}_{\text{TV}} + \alpha_3 \mathcal{L}_{\text{BVD}} + \alpha_4 \mathcal{L}_{\text{Reg}}$$

The α weights balance the loss terms; λ is reserved for the learned reconstruction mix.

MSE · accuracy

Fit the reference solution after K steps.

TV · oscillations

Penalize excess variation relative to the start.

BVD · interfaces

Penalize extra interface jumps beyond TENO5.

Reg · anchoring

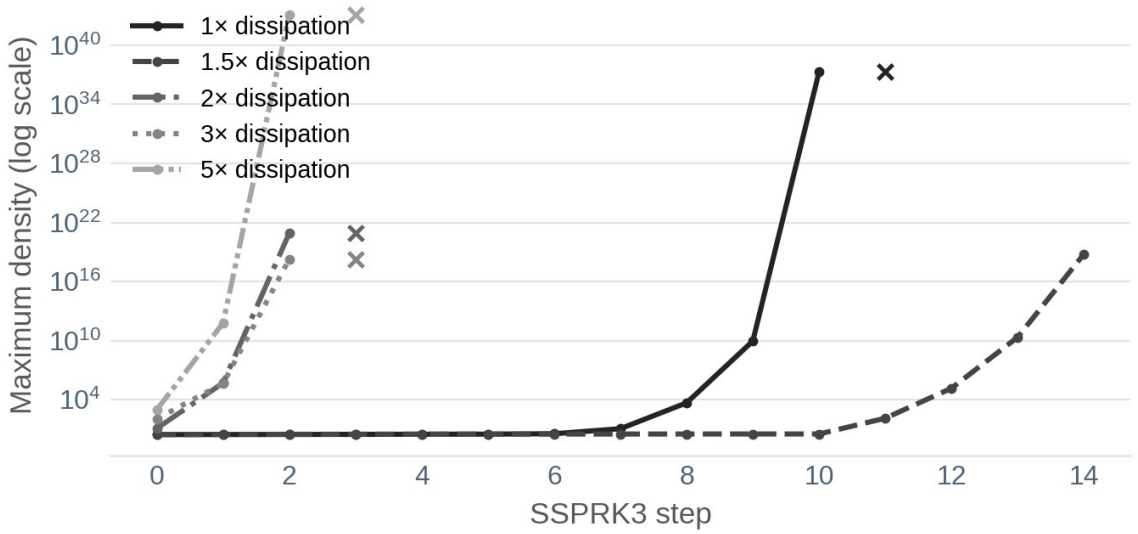
Keep weights close to the pretrained model.

Euler also adds a positivity penalty, measured before state repair; TV and Reg are normalized.

Euler instability shaped the training procedure

What failed

Historical training trace—not the final benchmark runs.



Invalid density or pressure destabilizes the Roe characteristic reconstruction.

What changed

01 Detect the real violation

Penalize invalid states before repair—not afterward.

02 Repair every SSPRK3 stage

Treat isolated violations; guard nonfinite states.

03 Normalize and bound training

Scale TV / Reg terms and clip large gradients.

04 Extend the horizon gradually

Warm-start the Euler curriculum: K = 2 → 4 → 8 → 15.

> Stable training enables evaluation; it does not by itself prove better accuracy.

What mixing solved—and what it did not

A small neural correction keeps the final reconstruction close to the classical baseline.

Imitation → **Physics training** → **Euler repairs** → **Learned mixing**

Separate two decisions

Classifier: which candidates to reject.

Mixing head: how much of the neural reconstruction to use.

Begin close to TENO5

The head starts near a 5% neural contribution.

A small scalar blend already helps; training improves it further.

Keep guarantees separate

Convex mixing is not a positivity or TVD proof.

State repairs are safeguards, not mathematical guarantees.

Final Sod and Shu–Osher runs remain finite and positive without invoking state repairs.

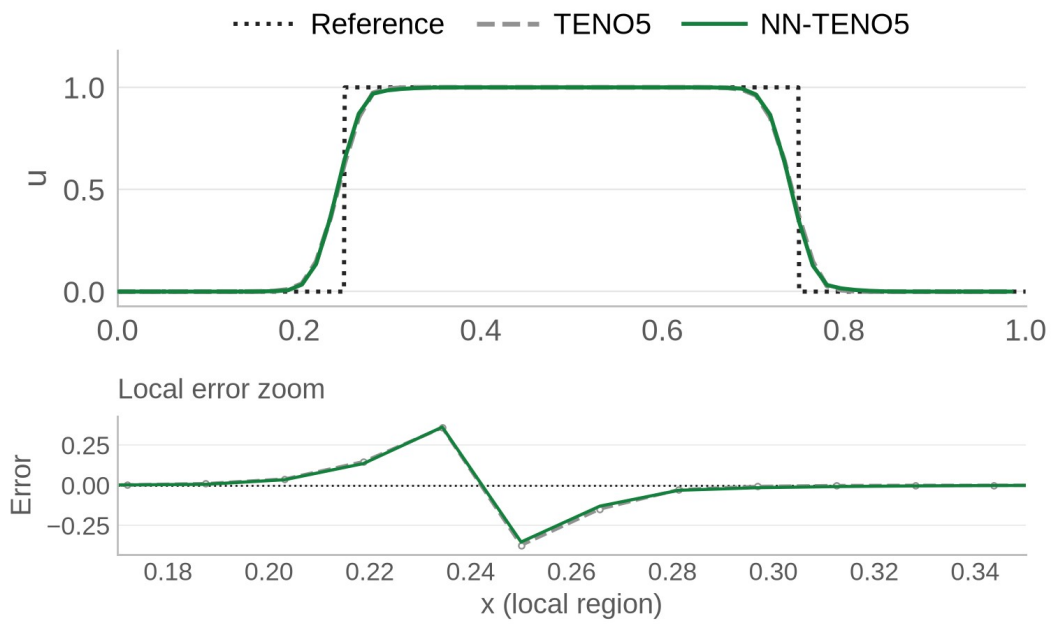
> Sod still has slightly higher error: better mixing alone has not removed the backbone's limit.

Final scalar models improve both benchmarks

Advection

3.5% lower L_2 error

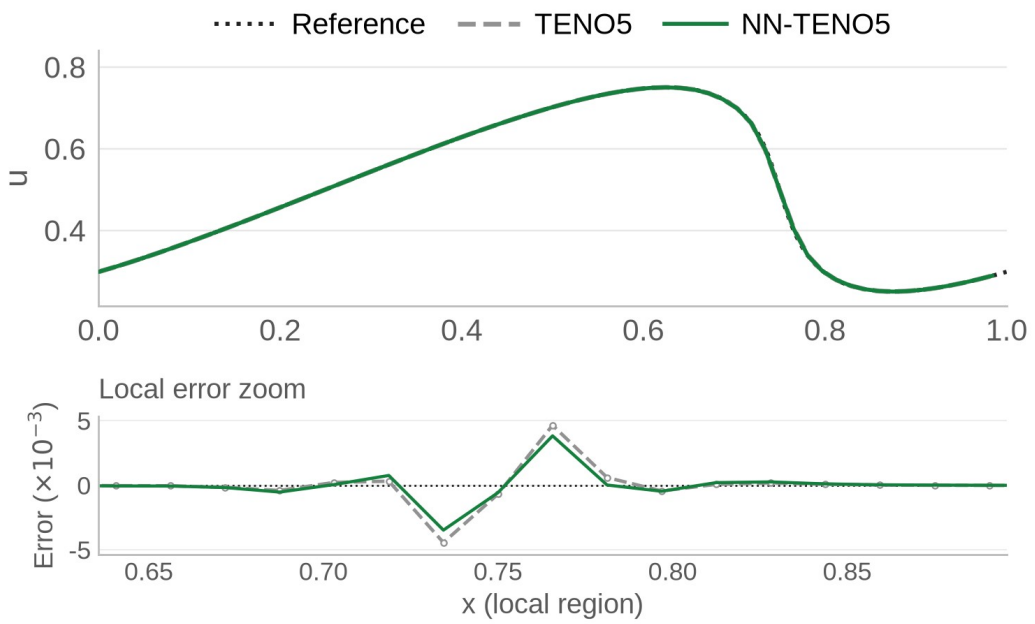
64 cells · CFL 0.5 · t = 1.0 · exact reference



Burgers

18.6% lower L_2 error

64 cells · CFL 0.5 · t = 0.5 · exact reference



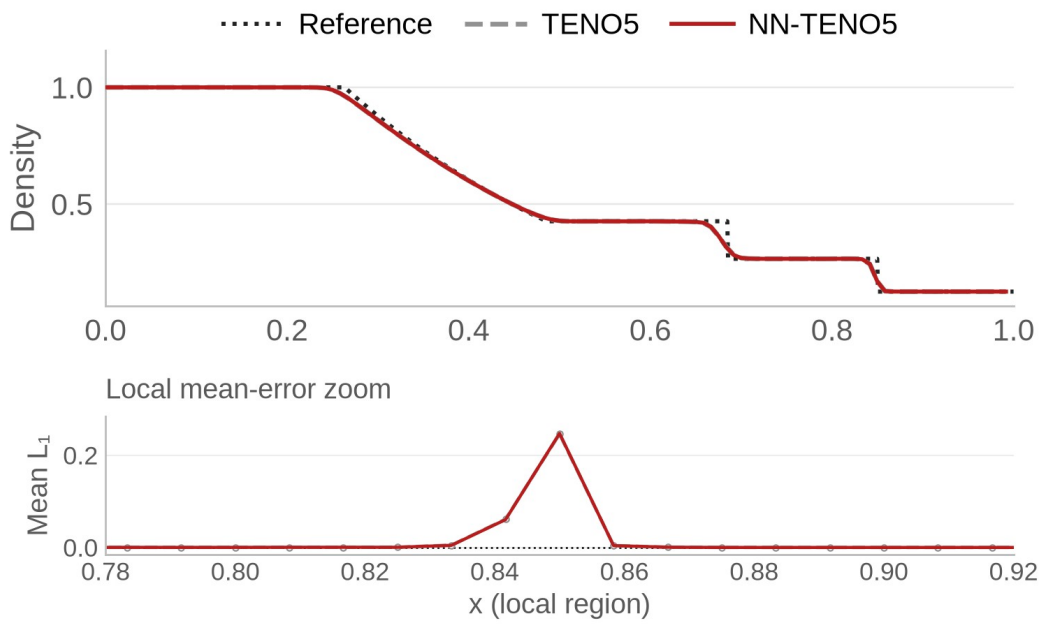
Burgers is evaluated before shock formation. Each lower panel shows a separate local-error zoom.

Euler gains depend on the benchmark

Sod

+0.35% error · slightly worse

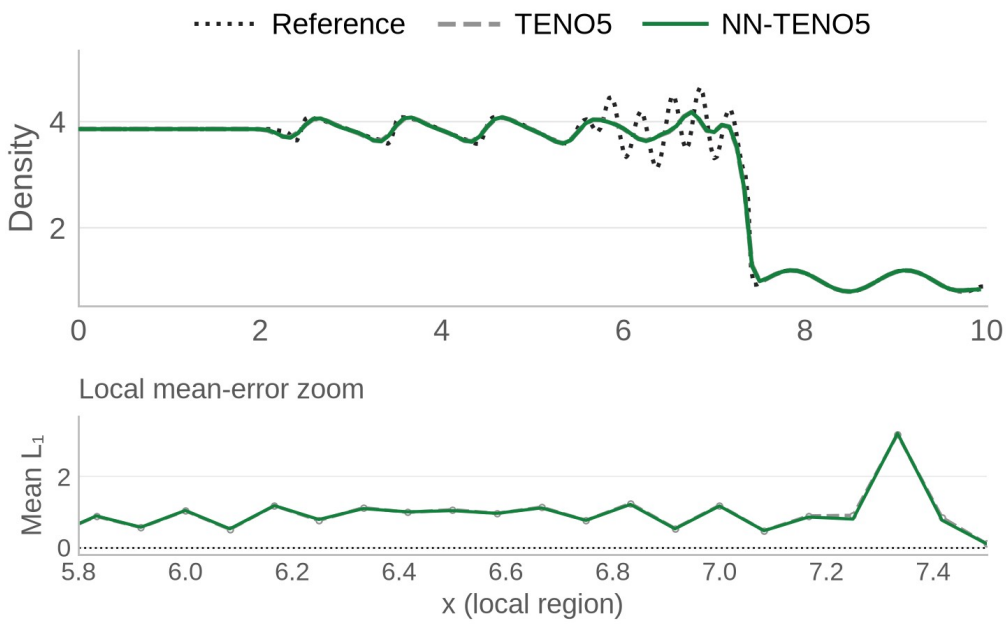
120 cells · t = 0.2 · exact Riemann reference



Shu–Osher

−1.84% error · modest improvement

120 cells · t = 1.8 · 320-cell TENO5 reference



CFL 0.4. Top: density. Local errors and percentages use mean L_1 over density, momentum and energy.

Canonical benchmarks quantify the final gains

Learned convex mixing compared with classical TENO5 on the same benchmark configuration.

BENCHMARK	ERROR METRIC	TENO5	NN-TENO5	RATIO R	CHANGE IN ERROR
Advection	Scalar L_2	9.8960×10^{-2}	9.5521×10^{-2}	0.9653	**-3.47%**
Burgers	Scalar L_2	8.1670×10^{-4}	6.6493×10^{-4}	0.8142	**-18.58%**
Sod	Mean Euler L_1	8.2583×10^{-3}	8.2871×10^{-3}	1.0035	**+0.35%**
Shu–Osher	Mean Euler L_1	2.1319×10^{-1}	2.0926×10^{-1}	0.9816	**-1.84%**

$$R = \frac{\mathrm{error}_{\mathrm{NN}}}{\mathrm{error}_{\mathrm{TENO5}}}, \qquad \mathrm{gain}(\%) = 100(1 - R)$$

Green: lower error. Red: higher error. Euler: mean L_1 over (ρ, ρv, E). Scalar: L_2.

Scope: four 1D cases at fixed grids—not a convergence study or a controlled runtime comparison.

A solver-integrated correction with measured limits

The contribution is the full tested workflow—not a classifier in isolation.

Differentiable reconstruction

TENO5 inside a differentiable conservative solver.

Train on evolved states using the four-term physics loss.

Euler diagnosis and repairs

Identify invalid-state failures.

Combine pre-repair penalties, substage safeguards and a rollout curriculum.

Dedicated learned mixing

Separate candidate decisions from their final influence.

A small mixing head retains the classical baseline.

Measured evidence **3 of 4 benchmarks improve** **Sod: +0.35% error**

Next: more grids and cases, backbone improvement, and a controlled accuracy–cost study.

No established convergence result, general accuracy guarantee, or formal positivity / TVD proof.

Selected references

Stevens & Colonius · 2020

Enhancement of shock-capturing methods via machine learning.

Theoretical and Computational Fluid Dynamics, 34, 483–496. DOI: 10.1007/s00162-020-00531-1

Fu, Hu & Adams · 2016

A family of high-order targeted ENO schemes for compressible-fluid simulations.

Journal of Computational Physics, 305, 333–359. DOI: 10.1016/j.jcp.2015.10.037

Li, Fu & Adams · 2022

A six-point neuron-based ENO (NENO6) scheme for compressible fluid dynamics.

arXiv preprint 2207.08500. arXiv: 2207.08500

Related foundations: NENO6 uses six stencil points; the model presented here uses five.

Thank you

Questions & discussion